

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q001 **Written**

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

Q002 Written**Find the average rate of change for the three given Try functions on their stated intervals.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q003 Written

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q005 **Written**

Find the given limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Q006 Written**Find the three given Try limits, including polynomial and h-quotient forms.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q007 **Written**

Find all nine given Exercise limits, including x-limits and h-limits.

Q008 Written

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Q010 Written

Find the differential coefficients at the given values for the two given Try functions using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q011 Written

Find the differential coefficients for all six given Exercise functions at their stated values.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Q014 Written**Differentiate the three given Try functions using the definition.**

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q015 **Written****Differentiate all nine given Exercise functions using the definition.**

Q016 Written

A regular hexagon has side length x . Write its area and differentiate it using the definition.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

Q018 MCQ

The average rate of change from a to b is:**Choose the correct option.**

A $[f(a)+f(b)]/(b-a)$

B $[f(b)-f(a)]/(b-a)$

C $f'(a)$

D $b-a$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q020 MCQ**The average area-rate for diameter 10 to 18 is:****Choose the correct option.**A 7π B 14π C 28π D 56π

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q021 MCQ

The h-form average rate is found by:**Choose the correct option.**

- A setting $h = 1$
- B cancelling the common h
- C differentiating first
- D squaring h

Q022 MCQ**A direct-substitution limit is valid when:****Choose the correct option.**

- A the expression is defined
- B the denominator is zero
- C the numerator is infinite
- D h is negative

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q023 MCQ

The 0/0 form requires:**Choose the correct option.**

- A factor/cancel first
- B differentiate twice
- C ignore the denominator
- D change the base

Q024 MCQ

In the finite-limit challenge, k equals:**Choose the correct option.**

A 3

B 5

C 6

D 9

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q025 MCQ

For a polynomial, the limit equals:**Choose the correct option.**

A its value at the point

B zero

C infinity

D the slope

Q026 MCQ

The definition of the differential coefficient at a is:**Choose the correct option.**

A $\lim [f(a+h)-f(a)]/h$

B $f(a+h)+f(a)$

C $[f(a)-f(0)]/a$

D $h/f(a)$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:**Choose the correct option.**

A -12

B -6

C 6

D 12

Q028 MCQ

The stone's velocity at 10 s is:**Choose the correct option.**

- A 49
 - B 98
 - C 196
 - D 4.9
-
-
-

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q029 MCQ

After cancelling h , take the limit as:**Choose the correct option.**A $h \rightarrow 0$ B $h \rightarrow 1$ C $x \rightarrow 0$ D $a \rightarrow \infty$

Q030 MCQ**The derivative function is:****Choose the correct option.**

- A the coefficient at one point
- B the differential coefficient for every x
- C the average rate only
- D the original function

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:**Choose the correct option.**A $x+3$ B $2x+3$ C $2x$ D x^2

Q032 MCQ

The area of a regular hexagon of side x is:**Choose the correct option.**

- A $3\sqrt{3}x$
 - B $(3\sqrt{3}/2)x^2$
 - C $6x^2$
 - D $\sqrt{3}x$
-
-
-

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q033 MCQ

A function need not be defined at a for its limit to:**Choose the correct option.**

- A exist
- B be infinite only
- C equal zero only
- D fail

Q034 MCQ

The derivative of a constant function is:**Choose the correct option.**

A 0

B 1

C x

D the constant

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Find the average rate of change of x^2 from 1 to 3.

Choose the correct option.

A 4

B 1

C 2

D 3

Quiz MCQ 2 Concept Quiz · MCQ**Find** $\lim_{x \rightarrow 2}(x^2 + 1)$.**Choose the correct option.****A** 0**B** 5**C** 2**D** 3

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Choose the correct option.

A 0

B 1

C $2x$

D 3

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s(t) = 4.9t^2$, find the velocity at $t=5$.

Choose the correct option.

A 0

B 1

C 2

D 49

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 5 Concept Quiz · Written

Find the average rate of change of $3x + 1$ from 2 to 7.

Quiz WRITTEN 6 Concept Quiz · Written**Evaluate** $\lim_{h \rightarrow 0} (6 + 2h)$.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

Quiz WRITTEN 8 Concept Quiz · Written

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

Answer key

Use the question number shown on each page.

Q001 4; $8 + h$	Q009 $f'(-2) = -12$	Q017 Yes; the limit is 5.	Q030 B
Q002 2; $-5/3$; $h - 6$	Q010 Substitute a into $[f(a + h) - f(a)]/h$	Q018 B	Q031 B
Q003 Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.	cancel h , and let h tend to zero.	Q019 C	Q032 B
Q004 $7\pi \text{ cm}^2/\text{cm}$	Q011 Use the definition at each specified point and simplify exactly.	Q020 A	Q033 A
Q005 -2 ; 6	Q012 98 m s^{-1}	Q021 B	Q034 A
Q006 Substitute after removing any common factor; polynomial limits are obtained directly.	Q013 $f'(x) = 2x + 3$	Q022 A	Quiz MCQ 1 A
Q007 Each limit is obtained by direct substitution or cancellation of the factor causing $0/0$.	Q014 Expand each $f(x + h) - f(x)$, cancel h and take the limit.	Q023 A	Quiz MCQ 2 B
Q008 6	Q015 Apply $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)]/h$ to every printed polynomial.	Q024 C	Quiz MCQ 3 C
	Q016 $A = \frac{3\sqrt{3}}{2}x^2$, $A'(x) = 3\sqrt{3}x$	Q025 A	Quiz MCQ 4 D
		Q026 A	Quiz WRITTEN 5 3
		Q027 A	Quiz WRITTEN 6 6
		Q028 B	Quiz WRITTEN 7 12
		Q029 A	Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Classified Practice

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Classified Practice

Differentiation and Tangent Lines

Q003 **Written**

Differentiate every given Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Classified Practice

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.

Q007 Written

Find the four quadratics satisfying the four given triples of derivative and function conditions.

Classified Practice

Differentiation and Tangent Lines

Q008 Written

Find the quadratic $f(x)$ satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

Q009 Written**Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.**

Classified Practice

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

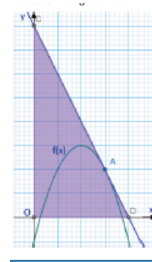
Q011 Written**Find the tangent line for all nine given Exercise functions at their given points.**

Classified Practice

Differentiation and Tangent Lines

Q012 **Written**

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Classified Practice

Differentiation and Tangent Lines

Q013 Written

Solve the two given Warm Up demands: tangents from $(C(0,2))$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Q014 Written

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Classified Practice

Differentiation and Tangent Lines

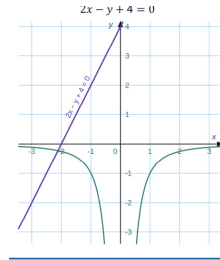
Q015 **Written****Solve all six given** Exercise demands involving external points or prescribed gradients.

Classified Practice

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Classified Practice

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Choose the correct option.**

A x^{n-1}

B nx^{n-1}

C $n + x$

D x^n

Q018 MCQ**The derivative of $V = \pi r^2 h$ with respect to r is:****Choose the correct option.**A πh B $2\pi r h$ C $2\pi r$ D πr^2

Classified Practice

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:**Choose the correct option.**

A -6

B -3

C 0

D 3

Q020 MCQ**The challenge function can be written as:****Choose the correct option.**

A $x^{5/3} + x^{-2}$

B $x^5 + x^2$

C $x^{3/5} + x^2$

D $x^{5/3} + x^2$

Classified Practice

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:

Choose the correct option.

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

Q022 MCQ**The condition $f'(0) = -2$ gives:****Choose the correct option.**

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

Classified Practice

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Choose the correct option.**

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

Q024 MCQ**The challenge coefficient a is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 6

Classified Practice

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Choose the correct option.**A $f(a)$ B a C $f'(a)$ D $1/f'(a)$

Q026 MCQ**The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:****Choose the correct option.**

A $y = x - 5$

B $y = 2x - 8$

C $y = -2x + 4$

D $y = 3x - 11$

Classified Practice

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:**Choose the correct option.**

A 2

B 3

C 4

D 8

Q028 MCQ**The area of triangle (ODC) in the challenge is:****Choose the correct option.****A** 8**B** 12**C** 16**D** 32

Classified Practice

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Choose the correct option.**A $((0,0))$ B $((a,f(a)))$ C $((f(a),a))$ D $((1,f'(a)))$

Q030 MCQ**The gradient of $2x - y + 4 = 0$ is:****Choose the correct option.****A** -2**B** -1**C** 1**D** 2

Classified Practice

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:

Choose the correct option.

A $-2/x^3$

B $2/x^3$

C $1/x^2$

D $-1/x^3$

Q032 MCQ**The parallel tangent in the challenge is:****Choose the correct option.**

A $y = 2x + 3$

B $y = 2x - 3$

C $y = -2x - 3$

D $y = x - 2$

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.

Quiz WRITTEN 2 Concept Quiz · Written

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at $((2,4))$.

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

The derivative of a constant is:

Choose the correct option.

A 0

B 1

C x

D the constant

Quiz MCQ 6 Concept Quiz · MCQ

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Choose the correct option.

- A three function values
- B two derivatives and one function value
- C one derivative only
- D the tangent gradient only

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 7 **Concept Quiz · MCQ**

The tangent equation in point-slope form is:

Choose the correct option.

A $y - y_1 = m(x - x_1)$

B $y = mx + c$

C $y + y_1 = m(x + x_1)$

D $x - x_1 = m(y - y_1)$

Quiz MCQ 8 Concept Quiz · MCQ

A tangent parallel to $y = 3x - 1$ has gradient:

Choose the correct option.

A -3

B $1/3$

C 3

D 1

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q008 $f(x) = 3x^2 - 18x + 27$

Q019 C

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q009 $y = 2x - 8$

Q020 A

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q021 B

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

Q022 B

Q012 16 square units

Q023 A

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q024 C

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q025 C

Q004 $\frac{10}{3}$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q026 B

Q005 $f(x) = x^2 - 2x + 3$

Q016 $y = 2x - 3$

Q027 C

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q017 B

Q028 C

Q018 B

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

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Formula opener

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Rate

$$\frac{dy}{dx}$$

Eng Eslam Ahmed | 01120009622

Classified Practice

Applications of Differentiation: Polynomial Functions

Q001 **Written**

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Q002 Written**Find the extreme values and sketch the three given Try cubics.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q003 **Written**

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q005 Written

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Q006 Written**Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q007 **Written**

Find the maximum and minimum values of all six given cubics on their given closed intervals.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q009 **Written**

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b , and the local minimum.

Q010 Written

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q011 Written

Solve all six given Exercise problems asking for cubic coefficients from extrema.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c .

Classified Practice

Applications of Differentiation: Polynomial Functions

Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

Q014 Written

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q015 Written

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q017 Written

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

Q018 Written

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q019 **Written****Find the parameter ranges in all six given Exercise problems for monotonic increase.**

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k + 6)x + 10$, find the k -range for monotonic increase on $x \in [1, \infty)$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q021 Written

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Q022 Written**Find the number of distinct real roots for the three given Try equations.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q023 **Written****Find the number of distinct real roots for all nine given cubic equations.**

Q024 Written**Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

Q026 Written**Solve the two given Try parameter problems for three or two distinct real roots.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q027 **Written****Solve all four given Exercise parameter problems for the required number of distinct roots.**

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q029 **Written**

Prove the given inequalities using an auxiliary cubic and a derivative sign table.

Q030 Written**Prove all three given cubic inequalities on their stated domains.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Q032 Written**Find the extreme values and sketch the two given quartic functions.**

Classified Practice

Applications of Differentiation: Polynomial Functions

Q033 Written

Find the extreme values and graphs for all six given quartic functions.

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q035 MCQ

At a local maximum, the sign of f' changes:**Choose the correct option.**A $- \rightarrow +$ B $+ \rightarrow -$ C $+ \rightarrow +$ D $- \rightarrow -$

Q036 MCQ**For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:****Choose the correct option.****A** $a - 1$ **B** a **C** $a + 1$ **D** $1 - a$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q037 MCQ

Absolute extrema on a closed interval require comparing:**Choose the correct option.**

- A stationary points only
- B endpoints only
- C stationary points and endpoints
- D the y-intercept

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:**Choose the correct option.**A $4\sqrt{2}$

B 9

C 6

D 0

Classified Practice

Applications of Differentiation: Polynomial Functions

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Choose the correct option.

A 2

B 6

C 12

D -12

Q040 MCQ**The derivative roots in the challenge are:****Choose the correct option.**A $-4, 2$ B $-2, 4$ C $2, 4$ D $-3, 3$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:**Choose the correct option.**A $D > 0$ B $D = 1$ C $D \leq 0$ D $D < 0$ only

Q042 MCQ

The challenge range for both extrema is:**Choose the correct option.**

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Choose the correct option.

- A negative everywhere
- B non-negative everywhere
- C zero everywhere
- D linear

Q044 MCQ

The challenge domain in 7-13 is:**Choose the correct option.**A $(-\infty, \infty)$ B $[1, \infty)$ C $[-1, 1]$ D $(0, 1)$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q045 MCQ

Three distinct roots occur when the x-axis lies:**Choose the correct option.**

- A above both extrema
- B between the local maximum and minimum
- C at an endpoint only
- D at the local maximum only

Q046 MCQ**The 7-14 challenge has how many distinct real roots?****Choose the correct option.****A** 0**B** 1**C** 2**D** 3

Classified Practice

Applications of Differentiation: Polynomial Functions

Q047 MCQ

Three intersections with $y = a$ require the level to lie:**Choose the correct option.**

- A strictly between the extrema
- B above the maximum
- C below the minimum
- D at an extremum

Q048 MCQ

The 7-15 warm-up range is:**Choose the correct option.**

A $-27 < a < 5$

B $-5 < a < 27$

C $a \leq -27$

D $a \geq 5$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q049 MCQ

To prove $\text{LHS} \geq \text{RHS}$, define:**Choose the correct option.**

A $f = \text{LHS} + \text{RHS}$

B $f = \text{LHS} - \text{RHS}$

C $f = \text{RHS} - 1$

D $f = f'$

Q050 MCQ

The largest k in the challenge is:**Choose the correct option.**

A $2 - 4\sqrt{2}$

B $4\sqrt{2} - 2$

C $2 + 4\sqrt{2}$

D $-2 - 4\sqrt{2}$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q051 MCQ

A repeated root of a derivative may be:**Choose the correct option.**

- A always a local maximum
- B always a local minimum
- C stationary but not an extreme
- D impossible

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:**Choose the correct option.**A $k = 27$ B $k < 27$ C $k > 27$ D $k \leq 0$ only

Classified Practice

Applications of Differentiation: Polynomial Functions

Q053 MCQ

A local minimum is identified by the change:**Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Q054 MCQ**For $y = x^3 - 6x^2 + 16$, the critical x-values are:****Choose the correct option.****A** 0, 4**B** 1, 3**C** -2, 2**D** 2, 4

Classified Practice

Applications of Differentiation: Polynomial Functions

Q055 MCQ

The derivative sign table is used to determine:**Choose the correct option.**

- A only the y-intercept
- B increase/decrease and extrema
- C the constant term
- D the domain

Q056 MCQ**On a closed interval, an endpoint can be:****Choose the correct option.**

- A ignored
- B an absolute extreme
- C a derivative root only
- D never a minimum

Classified Practice

Applications of Differentiation: Polynomial Functions

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Choose the correct option.

A $-2, 2$

B $-1, 1$

C $0, 4$

D $2, 4$

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Choose the correct option.

A $x = -3$

B $x = -\sqrt{2}$

C $x = \sqrt{2}$

D $x = 3$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Choose the correct option.

A $f(a) = 0$

B $f'(a) = 0$

C $f''(a) = 0$

D $a = 0$

Q060 MCQ**In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:****Choose the correct option.**

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Choose the correct option.

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

Q062 MCQ

Two distinct stationary points occur when the derivative discriminant is:**Choose the correct option.**

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Choose the correct option.

A non-negative for all x

B negative for all x

C linear

D undefined

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:**Choose the correct option.**A $[1, \infty)$ B $[-1, 1]$ C $(-\infty, 0)$ D $[0, 1]$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q065 MCQ

A cubic with both stationary values above the x-axis has:**Choose the correct option.**

- A three roots
- B one distinct real root
- C no real roots
- D two repeated roots

Q066 MCQ**To count cubic roots, compare the extrema with:****Choose the correct option.**

- A the y-axis
- B the x-axis
- C the derivative axis
- D the domain only

Classified Practice

Applications of Differentiation: Polynomial Functions

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:**Choose the correct option.**

- A one distinct intersection
- B two distinct intersections
- C three distinct intersections
- D no intersection

Q068 MCQ**For $x^4 - 4x^3 + k$, the global minimum occurs at:****Choose the correct option.****A** $x = 0$ **B** $x = 1$ **C** $x = 3$ **D** $x = 4$

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Quiz WRITTEN 2 Concept Quiz · Written

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 3 Concept Quiz · Written**Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.**

Quiz WRITTEN 4 Concept Quiz · Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)**At a local minimum, f' changes:****Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Quiz MCQ 6 Concept Quiz · MCQ

A cubic with a positive-leading derivative and $D < 0$ is:

Choose the correct option.

- A always increasing
- B always decreasing
- C constant
- D undefined

Classified Practice

Applications of Differentiation: Polynomial Functions

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

The 7-15 challenge requires the level to satisfy:

Choose the correct option.

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

Quiz MCQ 8 Concept Quiz · MCQ

The global minimum of $x^4 - 4x^3 + k$ is:

Choose the correct option.

A $k - 27$

B $k + 27$

C $k - 3$

D $27 - k$

Answer key

Use the question number shown on each page.

- Q001** Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.
- Q002** Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.
- Q003** Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.
- Q004** Local maximum at $x = a - 1$; local minimum at $x = a + 1$.
- Q005** Maximum 21 at $x = 2$; minimum -11 at $x = -2$.
- Q006** Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.
- Q007** For each function, compare the endpoint values with the values at all critical x-values lying in the interval.
- Q008** Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.
- Q009** $a = 12, b = -7$; local minimum -23 at $x = -2$.
- Q010** Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.
- Q011** Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.
- Q012** $a = -3, b = -24, c = 150$.
- Q013** $0 \leq k \leq \frac{1}{3}$.
- Q014** Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.
- Q015** Apply the derivative-discriminant criterion separately to each printed cubic.
- Q016** $k < 0$ or $k > 4$.
- Q017** $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.
- Q018** Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.
- Q019** Differentiate each cubic and impose that its quadratic derivative never becomes negative.
- Q020** $k \leq 3$.
- Q021** Three distinct real roots.
- Q022** Use the cubic graph: count x-axis intersections from the stationary points and end behavior.
- Q023** Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.
- Q024** One distinct real root.
- Q025** $-27 < a < 5$.
- Q026** Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .
- Q027** Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.
- Q028** $\frac{7-\sqrt{65}}{4} < a < \frac{7-\sqrt{33}}{4}$ or $\frac{7+\sqrt{33}}{4} < a < \frac{7+\sqrt{65}}{4}$.
- Q029** Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.
- Q030** Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.
- Q031** $k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.
- Q032** For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.
- Q033** Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.
- Q034** $k < 27$.
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** C
- Q040** B
- Q041** C
- Q042** B
- Q043** B
- Q044** B
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** B
- Q050** A
- Q051** C
- Q052** B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at
 $x = 2$

Quiz WRITTEN 2 $a = 3, b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

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Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Classified Practice

Integration: Definite Integrals and Applications

Q001 **Written****Find the two indefinite integrals in the Warm Up.**

Q002 Written

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Classified Practice

Integration: Definite Integrals and Applications

Q003 **Written****Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).**

Q004 Written

Solve the remaining eight given integrals, including products and squared brackets.

Classified Practice

Integration: Definite Integrals and Applications

Q005 Written

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Q006 Written**Find the two given** Try functions from their derivatives and one given value of F .

Classified Practice

Integration: Definite Integrals and Applications

Q007 **Written****Find $F(x)$ for all six given Exercise conditions.**

Q008 Written**Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.**

Classified Practice

Integration: Definite Integrals and Applications

Q009 Written

Solve the given Try tangent-gradient problem through $(1, 1)$.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.

Classified Practice

Integration: Definite Integrals and Applications

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Classified Practice

Integration: Definite Integrals and Applications

Q013 Written

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

Q014 Written

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Classified Practice

Integration: Definite Integrals and Applications

Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Q016 Written**Solve the four given Try definite-integral transformations.**

Classified Practice

Integration: Definite Integrals and Applications

Q017 **Written****Solve the first six given Exercise questions using interval splitting and reversal.**

Q018 Written**Solve the final six given Exercise questions using the factorized definite-integral identity.**

Classified Practice

Integration: Definite Integrals and Applications

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Q020 Written**Solve the two given Try integral equations for $f(x)$.**

Classified Practice

Integration: Definite Integrals and Applications

Q021 Written

Find $f(x)$ for all five given integral equations.

Q022 Written**Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .**

Classified Practice

Integration: Definite Integrals and Applications

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Q024 Written**Differentiate the three given expressions with respect to x .**

Classified Practice

Integration: Definite Integrals and Applications

Q025 Written

Find $f(x)$ and a for the three given equations with a variable upper limit.

Q026 MCQ**The constant of integration is required because:****Choose the correct option.**

- A every primitive is unique
- B primitives differ by a constant
- C the integrand is constant
- D the limits are equal

Classified Practice

Integration: Definite Integrals and Applications

Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Choose the correct option.**

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

Q028 MCQ**Before integrating $(x - 1)^2$, first:****Choose the correct option.**

- A differentiate it
- B expand it
- C set $x = 0$
- D add limits

Classified Practice

Integration: Definite Integrals and Applications

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Choose the correct option.**

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

Q030 MCQ**To find $F(x)$ from $F'(x)$, you should first:****Choose the correct option.**

- A differentiate again
- B integrate $F'(x)$
- C set $F = 0$
- D ignore the initial value

Classified Practice

Integration: Definite Integrals and Applications

Q031 MCQ

The condition $F(1) = 7$ determines:**Choose the correct option.**

- A the derivative
- B the domain
- C the constant C
- D the leading coefficient

Q032 MCQ**If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:****Choose the correct option.**

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

Classified Practice

Integration: Definite Integrals and Applications

Q033 MCQ

The gradient of the tangent at (x, y) is:**Choose the correct option.**A $f(x)$ B $f'(x)$ C $f''(x)$ D $x + y$

Q034 MCQ**A point (x_0, y_0) on $y = f(x)$ gives:****Choose the correct option.**

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

Classified Practice

Integration: Definite Integrals and Applications

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Choose the correct option.

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Choose the correct option.

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

Classified Practice

Integration: Definite Integrals and Applications

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Choose the correct option.

A 1

B a

C 0

D $f(a)$

Q038 MCQ

A reversed definite integral satisfies:**Choose the correct option.**

- A $\int_a^b f = \int_b^a f$
 - B $\int_a^b f = -\int_b^a f$
 - C both are zero
 - D only if $f = 0$
-
-
-

Classified Practice

Integration: Definite Integrals and Applications

Q039 MCQ

To evaluate a definite integral of a product, first:**Choose the correct option.**

- A integrate each factor separately
- B expand the product
- C differentiate the product
- D change the limits

Q040 MCQ**Adjacent integrals with the same integrand can be:****Choose the correct option.**

- A multiplied
- B combined over the joined interval
- C set equal to zero
- D reversed only

Classified Practice

Integration: Definite Integrals and Applications

Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta)dx$, the factors must have:

Choose the correct option.

- A different roots
- B the displayed roots (a,b)
- C zero coefficients
- D equal limits

Q042 MCQ

The identity for the factored integral is:**Choose the correct option.**

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

Classified Practice

Integration: Definite Integrals and Applications

Q043 MCQ

A minus sign before an integral can be removed by:**Choose the correct option.**

- A squaring the integrand
- B reversing its limits
- C adding C
- D differentiating

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Choose the correct option.

- A a function of x
- B a constant
- C always zero
- D a derivative

Classified Practice

Integration: Definite Integrals and Applications

Q045 MCQ

The best first step for an integral equation is to let:**Choose the correct option.**

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Choose the correct option.

A $1/3$

B $-1/3$

C 0

D 1

Classified Practice

Integration: Definite Integrals and Applications

Q047 MCQ

 $\frac{d}{dx} \int_a^x f(t)dt$ equals:

Choose the correct option.

A $f(a)$ B $f(x)$ C $F(x)$

D 0

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:**Choose the correct option.**

- A the lower limit
- B the upper-limit variable
- C C
- D zero

Classified Practice

Integration: Definite Integrals and Applications

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Choose the correct option.

- A the derivative
- B the unknown lower limit a
- C the integrand degree
- D the upper limit

Q050 MCQ**For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:****Choose the correct option.**

A $2x^2 + 3x - 5$

B $2x^3 + 3x^2 - 5x$

C $3x^2 + 3x - 5$

D 0

Classified Practice

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Classified Practice

Integration: Definite Integrals and Applications

Quiz WRITTEN 3 Concept Quiz · Written**Evaluate** $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Classified Practice

Integration: Definite Integrals and Applications

Quiz MCQ 5 **Concept Quiz · MCQ****The derivative of $\int_0^x (3t^2 + 1)dt$ is:****Choose the correct option.**

A $3x^2 + 1$

B $x^3 + x$

C $6x$

D 0

Quiz MCQ 6 Concept Quiz · MCQ**A primitive of $5x^4$ is:****Choose the correct option.**

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

Classified Practice

Integration: Definite Integrals and Applications

Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Choose the correct option.

A $2x^2 + 2x + 3$

B $4x^2 + 2x + 3$

C $2x^2 + 2$

D $4x + 5$

Quiz MCQ 8 Concept Quiz · MCQ

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Choose the correct option.

A $\int_1^5 f(x)dx$

B $\int_1^3 f(x)dx$

C $\int_3^5 f(x)dx$

D 0

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.	Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.	Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.	Q036 B
Q002 Expand products or powers, collect like powers, then integrate every term.	Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.	Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.	Q037 C
Q003 Apply linearity and the power rule; retain the constant of integration.	Q014 Use linearity, expand products, then apply the endpoint-substitution rule.	Q022 $2x^2 + 3x - 5$.	Q038 B
Q004 Expand each product or square, simplify, and integrate term by term.	Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.	Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a .	Q039 B
Q005 $F(x) = x^3 + 4x^2 - x + 3$.	Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.	Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.	Q040 B
Q006 Integrate the derivative and use the given point to determine C .	Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.	Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a .	Q041 B
Q007 For each item, integrate $F'(x)$, then impose the stated value of F .	Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$.	Q026 B	Q042 B
Q008 $f(x) = x^3 - x^2 + x - 19$.	Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.	Q027 B	Q043 B
Q009 Integrate the given gradient and use the point on the curve to determine the constant.		Q028 B	Q044 B
Q010 Integrate the polynomial gradient and apply the given point condition in each case.		Q029 A	Q045 B
Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.		Q030 B	Q046 B
		Q031 C	Q047 B
		Q032 A	Q048 B
		Q033 B	Q049 B
		Q034 B	Q050 A
		Q035 B	Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$
			Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$
			Quiz WRITTEN 3 6
			Quiz WRITTEN 4 $f(x) = x^2 - 1/3$
			Quiz MCQ 5 A
			Quiz MCQ 6 A
			Quiz MCQ 7 A
			Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

English Student Edition • Focused formulas and guided practice

Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

Classified Practice

Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

Q002 Written**Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.**

Classified Practice

Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x -axis.

Q004 Written**Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Q006 Written**Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.**

Classified Practice

Applications of Integration to Geometry

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.

Q008 Written**Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.**

Classified Practice

Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Q010 Written**Find the area bounded by $y = -x^3 + 4x$ and the x -axis.**

Classified Practice

Applications of Integration to Geometry

Q011 Written

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

Q012 Written

Evaluate $\int_{-2}^1 |x + 1| dx$.

Classified Practice

Applications of Integration to Geometry

Q013 Written

Evaluate $\int_0^3 |x^2 - 4| dx$.

Q014 Written**Evaluate** $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Q016 Written**Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.**

Classified Practice

Applications of Integration to Geometry

Q017 Written

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

Classified Practice

Applications of Integration to Geometry

Q019 **Written**

Solve the **given** bisection exercises for the given parabolas and lines, using the required area ratio.

Q020 MCQ**When the graph is below the x-axis, area is:****Choose the correct option.**

- A the signed integral
- B the negative of the integral
- C always zero
- D undefined

Classified Practice

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:**Choose the correct option.**

- A the graph and sign
- B only the y-intercept
- C the font size
- D the derivative only

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:

Choose the correct option.

A $8/3$

B $16/3$

C $32/3$

D $64/3$

Classified Practice

Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:**Choose the correct option.**

- A lower-upper
- B upper-lower
- C product
- D derivative

Q024 MCQ**If curves cross inside the interval, area must be:****Choose the correct option.**

- A split at the crossing
- B ignored
- C doubled always
- D negative

Classified Practice

Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Q026 MCQ**For cubic area, the sign changes at:****Choose the correct option.**

- A every turning point
- B each x-intercept
- C the y-intercept only
- D the midpoint

Classified Practice

Applications of Integration to Geometry

Q027 MCQ

The area for $y = -x^3 + 4x$ is:**Choose the correct option.**

A 4

B 6

C 8

D 12

Q028 MCQ**A negative lobe contributes:****Choose the correct option.**

- A negative area
- B positive absolute area
- C no area
- D a derivative

Classified Practice

Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:**Choose the correct option.**

- A the roots
- B the vertex only
- C the y-axis
- D the tangent

Q030 MCQ

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:

Choose the correct option.

A 3

B 4

C 5

D 6

Classified Practice

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c , split the integral:**Choose the correct option.**A at c

B at 0 only

C never

D after doubling

Q032 MCQ**A tangent line at $x=a$ has slope:****Choose the correct option.**A $f(a)$ B $f'(a)$ C a^2

D the area

Classified Practice

Applications of Integration to Geometry

Q033 MCQ

For two tangent lines, split at their:**Choose the correct option.**

- A intersection x-coordinate
- B y-intercept only
- C vertex only
- D origin

Q034 MCQ

The area in the given parabola-tangent Warm Up is:**Choose the correct option.**A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 3

Classified Practice

Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:**Choose the correct option.**

- A the same area
- B the same slope
- C the same root
- D zero area

Q036 MCQ**For $y=2x-x^2$, the total area above the x-axis is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

Classified Practice

Applications of Integration to Geometry

Q037 MCQ

The line $y=kx$ must satisfy the stated:**Choose the correct option.**

- A area ratio
- B font ratio
- C root count
- D period

Q038 MCQ**In a bisection problem, the cut-off area is:****Choose the correct option.**

- A the total area
- B half the total area
- C twice the total area
- D zero

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 1 Concept Quiz · MCQ

The area under $y = -x^2 + 4$ and above the x -axis is:

Choose the correct option.

A $16/3$

B 8

C $32/3$

D $64/3$

Quiz MCQ 2 Concept Quiz · MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A $16/3$

B $32/3$

C $64/3$

D $9/2$

Classified Practice

Applications of Integration to Geometry

Quiz MCQ 3 Concept Quiz · MCQ

Evaluate $\int_{-2}^1 |x + 1| dx$:

Choose the correct option.

- A 2
- B $5/2$
- C 3
- D $7/2$

Quiz MCQ 4 Concept Quiz · MCQ

The area enclosed by $y = x^2$ and tangents through (3, 8) is:

Choose the correct option.

A $1/3$

B $1/2$

C $2/3$

D $4/3$

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

Classified Practice

Applications of Integration to Geometry

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$ Q003 $\frac{32}{3}$ Q004 $\frac{32}{3}$ Q005 $\frac{32}{3}$ Q006 $\frac{64}{3}$ Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$ Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$ Q016 $\frac{64}{3}$ Q017 $\frac{2}{3}$ Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$ Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$